

Article

Learning to Improve Operational Efficiency from Pose Error Estimation in Robotic Pollination

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Abstract: Autonomous pollination robots have been widely discussed in recent years. However, the accurate estimation of flower poses in complex agricultural environments remains a challenge. To this end, this work proposes the implementation of a transformer-based architecture to learn the translational and rotational errors between the pollination robot's end effector and the target object with the aim of enhancing robotic pollination efficiency in cross-breeding tasks. The contributions are as follows: (1) We have developed a transformer architecture model, equipped with two feedforward neural networks that directly regress the translational and rotational errors between the robot's end effector and the pollination target. (2) Additionally, we have designed a regression loss function that is guided by the translational and rotational errors between the robot's end effector and the pollination targets. This enables the robot arm to rapidly and accurately identify the pollination target from the current position. (3) Furthermore, we have designed a strategy to readily acquire a substantial number of training samples from eye-in-hand observation, which can be utilized as inputs for the model. Meanwhile, the translational and rotational errors identified in the end-manipulator Cartesian coordinate system are designated as loss targets simultaneously. This helps to optimize the training of the model. We conducted experiments on a realistic robotic pollination system. The results demonstrate that the proposed method outperforms the state-of-the-art method, in terms of both accuracy and efficiency.

Keywords: pollination robot; transformer; offset errors



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1. Introduction

Robot-assisted pollination has been a topic for over a decade [1]. It is becoming an increasingly important contributor to crop production in the agricultural industry, and it is providing solutions to pressing challenges, such as the decline in natural pollinators. Pollination robots are categorized into several types, to meet diverse agricultural needs. Autonomous robots, equipped with advanced vision and artificial intelligence, navigate and pollinate independently. Remotely controlled or semi-autonomous robots require human oversight, suitable for complex environments. Unmanned aerial vehicles (UAVs) efficiently pollinate large fields from above. These robotic systems are designed to mimic the pollination process traditionally performed by bees and other insects. Utilizing advanced sensors, vision systems, and precision manipulators, robotic pollinators identify and interact with flowers, depositing pollen with high accuracy and consistency.

Various studies have been discussed in the field of automated pollination with applications in the pollination of kiwifruit [2,3], bramble flowers [4], vanilla [5], and tomato flowers [6]. The pollination technique requires the physical transfer of pollen from the male to the pistils (the reproductive organs of the female flowers) [7]. To minimize pollen loss and ensure precise transfer to the stigmas, it is necessary to address the difficulties

of collecting pollen and the challenges of maintaining its activity [8]. Unfortunately, this requires considerable time and effort, which has not been carefully considered in the existing pollination approaches. The difficulties originate from the very small size of the pistils, as well as the challenge of accurately identifying their rotation and orientation.

To this end, this work proposes a methodology based on a transformer architecture that aims to enhance the operational efficiency of robotic pollination in cross-breeding tasks. The output layer of the model incorporates two separate feedforward neural networks, each designed to regress the translational and rotational discrepancies between the robot and the pollination target, respectively. Unlike traditional transformer loss functions, we have designed a novel loss function that is informed by the translational and rotational discrepancies between the robot's end effector and the pollination targets. This enables the robot arm to rapidly and accurately identify the pollination target from the current observation view. Meanwhile, we have also designed a strategy to collect large-scale training samples from eye-in-hand observation and to designate the translational and rotational errors identified within the end-manipulator Cartesian coordinate system as loss targets. The observed images in the current and target positions, combined with the translational and rotational errors between them, provide a large scale for training data, thereby enhancing the model training. Figure 1 shows the pollination robot used in this work.

The remainder of this work is organized as follows: the related works are outlined in Section 2; the detailed descriptions of the method are introduced in Section 3; Sections 4–6 present the experiments, discussion, and conclusion, respectively.

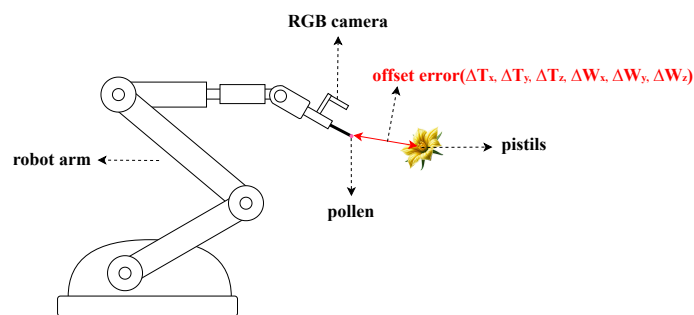


Figure 1. The pollination robot primarily consists of a UR5 robotic arm and a monocular RGB camera, configured in an eye-in-hand structure. At the end of the robotic arm, there is a pollination brush used for pollinating the pistils. The illustration depicts a robotic arm engaged in pollination, where the “offset error” marked in red represents the content that the model needs to learn. Here, ΔT_x , ΔT_y , and ΔT_z denote the components of the translational error along the X, Y, and Z axes, respectively, while ΔW_x , ΔW_y , and ΔW_z represent the components of the rotational vector, indicating the rotational errors.

2. Related Work

2.1. Pose Estimation

Achieving high precision in the end effector pose of a robotic arm poses significant challenges. To enhance the precision of the end effector pose, additional sensors, such as LiDAR or pressure sensors, are often employed, which can increase costs. Some methods are based on 6D object pose estimation, constructing templates to scan different positions in the input image, calculating similarity scores at each position, and obtaining the best match by comparing the similarity scores [9,10]. Others use feature-based methods, where CNN models directly regress the 3D coordinates of each pixel [11], or they describe the posterior density of a specific object pose through CNNs, comparing the observed image with the rendered image [12]. Additionally, some methods combine the advantages of template-based and feature-based approaches within a deep learning framework. The network integrates bottom-up pixel-level labeling with top-down object pose regression, predicting

the 3D position and 3D rotation of the object in a decoupled manner [13]. All these methods invariably use depth sensors to capture the depth information D of the target object. However, for scenarios where depth information cannot be captured, such as in the agricultural pollination robot discussed in this paper, the hollow regions of pollination flowers result in poor depth information acquisition. Moreover, the trained models incorporating depth information [11–13] do not directly regress and predict the 3D position information of the target object. The aforementioned template-based and feature-based models are trained on point clouds, but they still face challenges when handling non-convex polyhedra and non-manifold data. Existing point-cloud training methods are primarily suitable for regular shapes, such as convex polyhedra and manifolds. These methods are limited when dealing with the complex and irregular shapes of flowers in agricultural environments, which often present non-polyhedral and non-manifold structures. This limitation makes it difficult to directly apply traditional point-cloud training methods. Additionally, there is a discrepancy between point-cloud training in virtual environments and real-world applications, which can affect the model's performance in real-world scenarios.

By contrast to the pose-estimation techniques introduced earlier, which rely on additional sensors to capture depth-enhanced image data for constructing point-cloud training—a practice difficult to implement in non-convex and non-manifold agricultural environments—our method utilizes only conventional cameras, to acquire RGB images from the pollination robot's end effector. This approach enables seamless end-to-end training in real-world settings, thereby eliminating the discrepancies commonly found between point-cloud training in virtual environments and their practical application in the field.

2.2. Traditional and Deep Learning Methods for Improving Precision

To improve the precision of the pollination end effector, traditional methods can be realized through the design of the robotic arm. For example, LI K designed a robotic arm for kiwifruit pollination [14], achieving high precision by expressing the end effector velocity and joint angular velocities through the Jacobian matrix. During trajectory planning, quintic polynomial interpolation ensures the continuity and smoothness of the velocity and acceleration curves of each joint. Forward kinematics and Monte Carlo methods are used for simulation analysis, to cover the entire pollination area with the end effector. In the vision system, a binocular camera [15] is used to obtain the 3D coordinates of flowers in real time, combining with the control system for trajectory planning and interpolation point calculation, to precisely control each joint motor and achieve point-to-point precise pollination. However, this robotic arm relies on binocular cameras to obtain the 3D coordinates of the flowers, and strong light or occlusions can affect the accuracy of the vision system.

Traditional algorithms can still achieve certain effects in specific agricultural pollination domains. A recent study by N Duc Tai [16] involved using segmentation methods to locate and segment cantaloupe flowers, employing mathematical models based on the biological characteristics of the cantaloupe flowers to determine the key points of each flower's growth direction, and using an inverse-projection method to convert the position of the flower from a 2D image to a 3D space, thus achieving flower pose localization.

In recent years, with the development and application of deep learning, STRADER J's BrambleBee [4] pollination robot has combined traditional algorithms and deep learning algorithms to estimate flower poses. The system uses a two-stage image-processing framework to recognize flowers and estimate their poses. It first employs a Naive Bayes pixel-level segmentation algorithm, then uses a convolutional neural network for classification and pose refinement, ensuring the accuracy of the recognition results. Finally, an octree-based obstacle map and a factor graph represent the flower map, mapping the flowers and their positions in 3D to optimize the pose estimation. Khubaib Ahmad's automatic watermelon flower pollination robot [17] utilizes deep learning to detect the 2D information of flowers and combines it with traditional algorithms, to calculate the depth of the flower positions, thereby achieving flower localization. The robot then adjusts the mechanical arm, using servo searching, until it reaches the ideal pollination position.

YANG's [18] vision-based servo automated pollination robot uses more advanced target-detection algorithms, such as YOLOv5, YOLACT++ [19], and DETR [20], to detect flowers, identifying the position and orientation of the pistil through rotation object-detection technology. The system then employs a pseudo-binocular ranging strategy, calculating the 3D coordinates of the pistil by moving the camera position and using eye-hand calibration to transform these coordinates into the robot arm's operational coordinate system. During the pollination task, the system combines visual servo control strategies, performing coarse positioning by moving the end effector near the flower, followed by precise positioning using a circular search trajectory to ensure accurate contact between the pollen brush and the pistil. However, the system still has an end effector precision error of 15 mm. Although the circular search trajectory can eventually achieve precise positioning, the process is time-consuming, taking nearly 19 s to complete pollination of a single flower. If the end effector precision can be improved, the efficiency of precise positioning through the circular search trajectory will significantly benefit.

Both the traditional methods of N Duc Tai and the deep learning-based pollination robots of STRADER J, Khubaib Ahmad, and YANG significantly impact the success rate and efficiency of pollination operations through errors in estimating the pose of pollination targets. For instance, the pollination robots of N Duc Tai and YANG can only ensure successful pollination by maintaining a large servo search area, which leads to reduced pollination efficiency. Consequently, our work proposes a deep learning method to learn the translational and rotational errors between the flower and the pollination end effector. By compensating for these errors, the positioning accuracy of the robot's end effector is enhanced. This reduction in the servo mechanism's search range and path ultimately improves the efficiency of the pollination process.

2.3. Transformer Architecture and Position Encoding

The transformer architecture, introduced by Vaswani et al. [21], has significantly impacted Natural Language Processing (NLP) and computer vision. Transformers utilize the self-attention mechanism to capture complex dependencies within input data, making them effective for understanding visual information. In computer vision, transformers process images by segmenting them into patches and linearly embedding these segments, enabling powerful attention mechanisms for tasks like image classification, object detection, and generating image descriptions. In robotics, transformers can predict the error offsets of a robotic arm's end by analyzing sequences of images or sensor data, enhancing precision and accuracy.

The introduction of position encoding [22] in predicting translational offset error and rotational pose offset error at the robotic arm's end further improves spatial awareness and accuracy. By embedding spatial position information directly into the transformer model, it enhances the model's ability to discern the relative and absolute positions of objects in a scene, which is critical for precise error estimation.

The self-attention mechanism of transformers can capture long-range dependencies, allowing the model to directly calculate the relationships between any two elements within an input image feature sequence, considering all the input data rather than just local information. This characteristic helps reduce the issue of local optima. Additionally, through fine-grained attention allocation, transformer-based models can identify and emphasize the features and relationships most relevant to the current task. Therefore, employing a transformer model equipped with positional encoding to predict the translational and rotational errors between the pollination robot's end effector and the target object is a viable approach.

3. Method

3.1. Offset Error

To describe the translational offset error and the rotational pose offset error, two Cartesian coordinate systems, C_A and C_B , need to be constructed at the target flower and at

the end effector of the pollination robot, respectively. The Cartesian coordinate system C_A , constructed on the flower, has its origin O at the end of the pistil. The plane formed by the x and y axes is parallel to the plane of the flower petals, and the z axis is parallel to the pistil, pointing inward, as shown in Figure 2b,c. The Cartesian coordinate system C_B , constructed at the end of the pollination robot, has its origin O at the end of the brush, obtained by translating the UR5 robotic arm's TCP coordinate system, as shown in Figure 2d. C_A can be derived from C_B through a 4×4 translation–rotation matrix TR :

$$TR = \begin{bmatrix} R_{11} & R_{12} & R_{13} & t_1 \\ R_{21} & R_{22} & R_{23} & t_2 \\ R_{31} & R_{32} & R_{33} & t_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1)$$

$$C_A = TR \cdot C_B \quad (2)$$

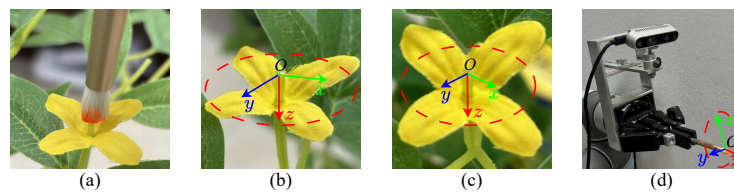


Figure 2. (a) The ideal zero-error pollination position. (b–d) The Cartesian coordinate systems' positions on the flower and at the end of the robotic arm, respectively.

The translational offset error and rotational pose offset error are the translation vector t and the rotational vector derived from the rotation matrix R of the translation–rotation matrix TR , respectively:

$$t = [t_1 \quad t_2 \quad t_3]^T \quad (3)$$

$$R = \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix} \quad (4)$$

When the end effector of the pollination robot is in the ideal pollination posture, the values of the translational offset error and the rotational pose offset error approach zero. At this moment, the brush at the end of the pollination robot is perpendicular to the plane of the petals of the target flower and just in contact with the pistil, i.e., the coordinate systems C_A and C_B coincide, as shown in Figure 2a.

$$\begin{aligned} \theta &= \cos^{-1} \left(\frac{\text{trace}(R) - 1}{2} \right) \\ \vec{V} &= \frac{1}{2 \sin(\theta)} \begin{bmatrix} R_{32} - R_{23} \\ R_{13} - R_{31} \\ R_{21} - R_{12} \end{bmatrix} \\ \vec{W} &= \theta \vec{V} \end{aligned} \quad (5)$$

The translational offset error and rotational pose offset error predicted by the model are denoted as $\hat{t}$ and $\hat{R}$, respectively. The differences between the actual and predicted translational offset error and rotational pose offset error from the end of the robotic arm to the pistil of the target flower are calculated as the translational error (TE) and the rotational error (RE), respectively. According to Equation (5), the rotation matrices R and $\hat{R}$ can be converted into rotational vectors $\vec{W}$ and $\vec{\hat{W}}$:

$$TE = \|\hat{t} - t\| \quad (6)$$

$$RE = 1 - \frac{\hat{\vec{W}} \vec{W}}{\|\hat{\vec{W}}\| \|\vec{W}\|} \quad (7)$$

The unit of TE is centimeters (cm), representing the spatial distance between the predicted translational error and the actual translational error. RE is dimensionless, representing the cosine distance between the predicted rotational vector error and the actual rotational vector error.

3.2. Attention Module

The attention module is implemented based on the self-attention mechanism, which enables the model to weigh the importance of different features within the input data. By applying this mechanism, the model can adaptively concentrate computational resources on regions of the image that contain key information for predicting pose offsets $(\Delta T_x, \Delta T_y, \Delta T_z, \Delta W_x, \Delta W_y, \Delta W_z)$. Furthermore, with the introduction of position encoding in the self-attention mechanism, the model achieves a comprehensive understanding of the image. This provides additional information for addressing symmetry issues in object pose prediction, enhancing the model's accuracy and augmenting its general capability in various scenarios encountered by pollination robots. This method not only improves the accuracy of the model but also strengthens its versatility in the diverse conditions faced by pollination robots.

3.3. Feature Extraction

Given the unique ability of transformer models to process image data, we utilize a pre-trained convolutional neural network, ResNet-50 [23], as the feature extraction network, to convert original images into high-dimensional feature vectors. These features are then fed into the transformer model for further processing. We compare feature extraction networks of different depths, to determine the optimal feature representation. This approach leverages the strong ability of ResNet-50 to capture detailed spatial hierarchies in images, providing a rich set of features for the transformer model to analyze. By incorporating this hybrid architecture, combining the strengths of CNNs in feature extraction with the advanced attention mechanism of transformers, the model achieves a nuanced understanding of image content relevant for predicting the pose offset errors. This method facilitates identification and focusing on crucial aspects of the input data, thereby enhancing the accuracy and efficiency of the pose-estimation process.

3.4. Proposed Approach

This work introduces a novel network model incorporating an attention module, designed to enhance prediction accuracy by focusing on the most critical parts of the input data. This model is particularly suited for analyzing image data captured by the end effector of a pollination robot, aiming to accurately predict the position offset of the end effector relative to the target object in a Cartesian coordinate system. During training, the model takes image data from the robotic arm's end as input, and it outputs a six-dimensional vector $(\Delta T_x, \Delta T_y, \Delta T_z, \Delta W_x, \Delta W_y, \Delta W_z)$ representing the translational offset errors $\Delta T = (\Delta T_x, \Delta T_y, \Delta T_z)$ and the rotational offset errors $\vec{W} = (\Delta W_x, \Delta W_y, \Delta W_z)$ in the three directions of the Cartesian coordinate system.

3.5. Offset Error Prediction

The input to the model is an RGB image captured by an "eye-in-hand" RGB camera on the robotic arm, depicting the positional state of the robotic arm's end effector during pollination and the pollination flower. The output is a vector representing the pose offset vector $(\Delta T_x, \Delta T_y, \Delta T_z, \Delta W_x, \Delta W_y, \Delta W_z)$, which includes the translational and rotational errors of the robot's end effector relative to the target object. To achieve this, the model employs two feedforward neural networks to directly map the attention-focused features

into the three-dimensional offset space and the three-dimensional rotational space. Figure 3 illustrates the entire algorithmic process.

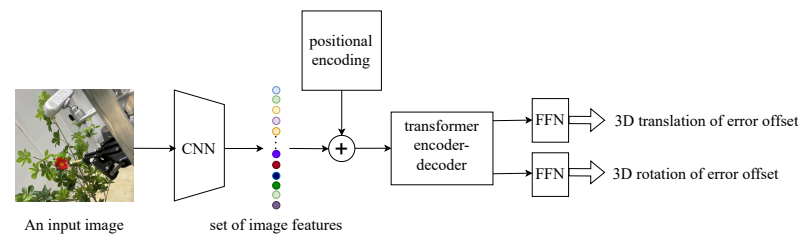


Figure 3. The diagram illustrates the workflow of the proposed algorithm. The process begins with the input image, which undergoes feature extraction via a convolutional neural network (CNN), to obtain high-dimensional image features. These features are then augmented with positional encoding before being input into the transformer module. The transformer’s output is further processed by two distinct feedforward neural networks, tasked with predicting translational and rotational errors, respectively.

3.6. Network Architecture

In this work, we have adopted a customized transformer model designed to process serialized image features and predict the translational offset error and rotational pose offset error at the end of a robotic arm. We have modified the original transformer model by introducing two-dimensional positional encoding, to preserve the spatial information of the input images. The output layer is customized, to generate 3D translational offset and a rotational pose offset error vector. As shown in Figure 4, a ResNet50 serves as the backbone for extracting a rich feature set from the input image data. To retain the spatial information among the elements in the images, we apply sinusoidal position encoding to the extracted features, which are then summed with the features before being fed into the encoder. To capture information from different subspaces, a multi-head attention mechanism with five heads is employed in the encoder. The output of the encoder is subsequently fed into the decoder. Following the standard architecture of the transformer, the decoder utilizes a multi-head attention mechanism to transform two embeddings of size d . These query objects are transformed by the decoder into output embeddings, which are then decoded by two separate feedforward networks into the translational offset error and the rotational pose offset error, respectively.

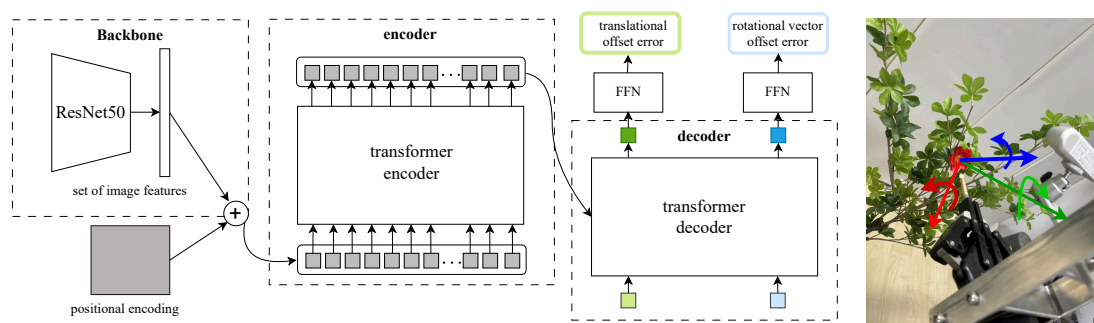


Figure 4. The final model employs ResNet50 as the backbone network to learn high-level features from the input image. These features are then supplemented with positional encoding before being passed to the encoder. Subsequently, the decoder first outputs a feature vector for the translational offset error, which is used as input for a feedforward neural network designed for prediction. This feature vector is then used as the query input for the decoder, resulting in another feature vector that is fed into a separate feedforward neural network for predicting the rotational pose offset error.

4. Experiments

4.1. Data Preprocessing

The image data used in this experiment are all from a proprietary dataset, which consists of images captured by a camera mounted on the robotic arm during the pollination process, where each image uniquely represents the pose of a flower to be pollinated. As shown in Figure 5, the orientation of the flowers relative to the end of the robotic arm was categorized into five directions: left, right, upwards, downwards, and front. Detailed statistical information is available in Table 1.

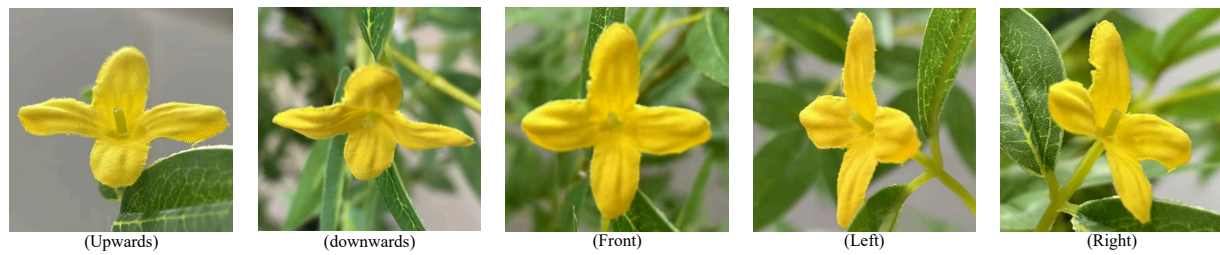


Figure 5. Flowers with different orientations relative to the end effector of the robotic arm.

Table 1. The dataset includes the number of images of flowers oriented in different directions relative to the the end effector of the robotic arm.

Orientations	Number of Flowers	Proportion (%)
L	261	20.6
R	274	21.7
F	230	18.2
U	256	20.3
D	243	19.2

"F", "L", "R", "U", and "D" represent front, left, right, upwards, and downwards orientations.

To ensure the consistency of the input data, all images captured by the robotic arm's end camera were first resized to a uniform resolution of (224×224) . Subsequently, the images were normalized, scaling the pixel values to the $[0, 1]$ range, to enhance the stability of the model training. Moreover, a series of data-augmentation techniques, including scaling, cropping, and color transformation, were applied, to increase data diversity and prevent overfitting. Regarding the label data, we assumed the position offset error $\Delta T = (\Delta T_x, \Delta T_y, \Delta T_z)$ and the rotational offset error $\Delta \vec{W} = (\Delta W_x, \Delta W_y, \Delta W_z)$, with $\|\Delta T\| < D$, where D was a constant. The position offset error Equation (8) and rotational offset error Equation (9) were normalized and scaled to $[-1, 1]$, as follows:

$$\Delta T_n = \frac{\Delta T}{D} \quad (8)$$

$$\vec{W} = \theta \vec{V} \quad (9)$$

$$\theta = \sqrt{\Delta W_x^2 + \Delta W_y^2 + \Delta W_z^2} \quad (10)$$

$$\vec{V} = \left(\frac{\Delta W_x}{\theta}, \frac{\Delta W_y}{\theta}, \frac{\Delta W_z}{\theta} \right) \quad (11)$$

The symbol θ in the equation (10) represents the rotation angle, and $\vec{V}$ denotes the unit vector along the axis of rotation. The normalized label data were $(\Delta T_n, \vec{V}, \frac{\theta}{2\pi})$.

4.2. Evaluation Metrics

In the model presented in this paper, we separately predict the translational offset error and the rotational pose offset error, thus designing two loss functions. The first loss

function, named $Loss_T$, measures the mean squared distance between the spatial position offset error predicted by the model for the end effector of the pollination robotic arm to the pistil of the target flower and the actual spatial offset error. $Loss_T$ is defined as follows:

$$Loss_T = \frac{1}{2m} \sum_{x \in \mathcal{M}} (\hat{T}_n - T_n)^2 \quad (12)$$

where $\mathcal{M}$ is the set of the test dataset, m is the number of elements in the set, and $\hat{T}_n$ and T_n are the translational offset error predicted by the model and the true translational offset error obtained via Equation (8), respectively.

The second loss function, named $Loss_R$, measures the discrepancy between the spatial rotational error predicted by the model for the end effector of the pollination robotic arm to the pistil of the target flower and the actual spatial rotational error. $Loss_R$ is defined as follows:

$$Loss_R = \frac{1}{2m} \sum_{x \in \mathcal{M}} \left(\frac{1}{\sigma_1^2} (\|\hat{\theta} - \theta\|^2) + \frac{1}{\sigma_2^2} \left(1 - \frac{\hat{\vec{V}} \cdot \vec{V}}{\|\hat{\vec{V}}\| \|\vec{V}\|} \right) + \log(\sigma_1 \sigma_2) \right) \quad (13)$$

where $\mathcal{M}$ is the set of the test dataset and m is the number of elements in the set. Variables $\hat{\theta}$, θ , $\hat{\vec{V}}$, and $\vec{V}$ represent the predicted and actual rotational angles and axes obtained via Equations (9)–(11), with σ_1 and σ_2 being the parameters that need to be learned.

The combined loss function used for model training is defined as

$$Loss = \alpha Loss_T + \beta Loss_R \quad (14)$$

This loss function in Equation (14) comprehensively measures the translational offset error loss and the rotational offset error loss during model training. Parameters α and β are hyperparameters representing weights that need to be systematically adjusted during model training.

4.3. Training Details

The model training was conducted in a computational environment equipped with NVIDIA V100 GPUs. We used the Adam optimizer [24], with the initial learning rate set to 0.01, and we employed a learning rate decay strategy that gradually reduced the learning rate to 0.00001 as training progressed. A custom loss function, Equation (14), was used during the training process. In the model training, the setting of the hyperparameters in Equation (14) affected the model's ability to converge. After extensive experiments, we finally set $\alpha = 0.0025$ and $\beta = 1$, which allowed the model to converge more easily during training.

In the dataset, flowers with different orientations were randomly shuffled, and then the data were divided into 10 subsets, using the K-fold cross-validation method. Each time, one subset was used as the test set, and the remaining nine subsets were used as the training set. This process was repeated 10 times. During the model training, we observed that the model initially converged rapidly and then gradually stabilized. Figure 6 shows the changes in the values of loss, translational error, and rotational error during the model training process:

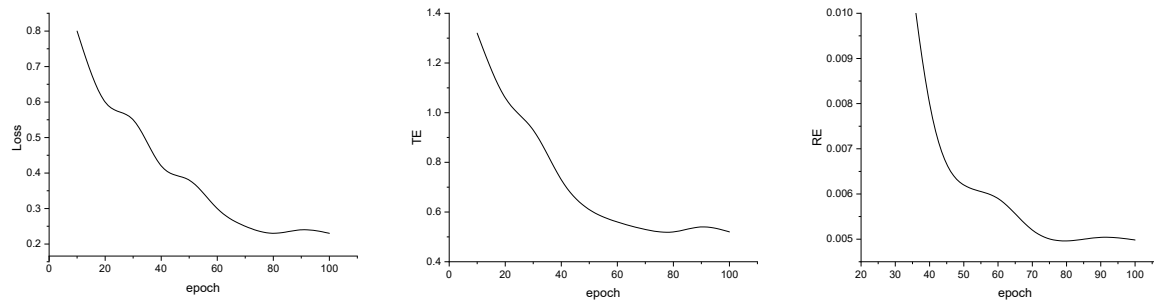


Figure 6. Changes in loss, TE , and RE during the model training process.

4.4. Results

To the best of our knowledge, we are the first to predict the translational offset error and rotational pose offset error of a robotic arm's end effector relative to a target object using only RGB image information. Therefore, this paper focuses on the accuracy improvements of the translational offset error and rotational pose offset error of the robot's end effector when using our proposed method compared to the state-of-the-art YANG method [18] as the baseline. Additionally, we highlight the enhancement in the efficiency of pollinating a single flower.

We conducted experiments on flowers with different orientations in groups, to analyze translational and rotational offset errors as well as detection speed. The experimental results showed slight variations for flowers with different orientations. As shown in Table 2 and Figure 7, flowers facing forward achieved the best results, in terms of experimental accuracy, with the smallest translational offset error and rotational pose offset error compared to flowers in other orientations. Flowers facing upwards came next. Due to environmental symmetry, there was almost no difference in results for flowers facing left and right. Flowers facing downwards had the worst results, in terms of both translational offset error and rotational pose offset error. However, the detection speed of the model for flowers with different orientations remained almost constant.

Table 2. Experimental results of the model on flowers with different orientations.

Orientations	TE	RE	FPS
L	0.82	0.0049	41
R	0.82	0.0049	42
F	0.78	0.0047	43
U	0.80	0.0048	42
D	0.87	0.0051	41

"F", "L", "R", "U", and "D" represent front, left, right, upwards, and downwards orientations.

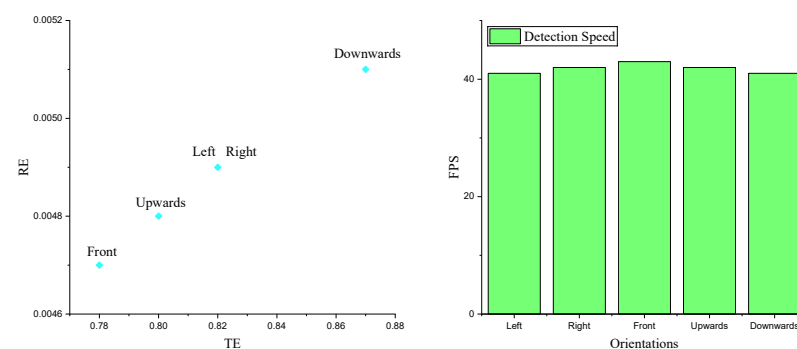


Figure 7. Distribution of the model's rotational and translational error detection for flowers with different orientations, as well as the detection speed.

We conducted our experiments using the YANG pollination robot. The YANG pollination robot's process of pollinating a single flower can be divided into five steps in

chronological order. After the fourth step, the positioning accuracy of the YANG pollination robot's end effector relative to the target pistil was 1.5 cm. At this point, we introduced a new step called "Fine adjustment position". We input the image of the flower's pose relative to the robot's end effector into our trained model, to predict the translational and rotational offset errors. Based on the predicted values, the pose of the pollination robot's end effector was adjusted.

As shown in Table 3, thanks to the new "Fine adjustment position" step, the translational distance and rotational discrepancy between the pollination robot's end effector and the pollination target were further reduced, narrowing the range for the next servo search pollination step. Our experiments showed that the average time for servo search pollination was only 3.1 s, achieving a pollination success rate comparable to YANG's pollination robot at 86.19%. As shown in Table 4, after applying our method to the pollination robot, the average distance accuracy between the robot's end effector and the target pistil reached 0.81 cm, an improvement of 46.67%. The rotational pose offset error calculated according to Equation (14) also reached 0.0049. The total time to complete the pollination of a single flower was reduced by nearly half, with an average efficiency improvement of 50.9%.

Table 3. Comparison of the average time cost for each step of the pollination system after incorporating our method.

Step	YANG (Baseline) Time Cost (S)	Our Time Cost (S)
Flower detection	0.0928	0.0927
Pistil identification	0.025	0.024
Position calculation (robot motion included)	1.8	1.8
Flower reaching (robot motion included)	4.2	4.2
Fine adjustment position	/	0.024
Servoing (robot motion included)	12.7	3.1

The "Fine adjustment position" step is an additional step. With this step included, the time spent in the "Servoing" step is greatly reduced, achieving the same pollination success rate in just 3.1 s. All numbers are recorded in seconds.

Table 4. Comparison of translational offset error and rotational pose offset error when applying our algorithm to flowers oriented in different directions compared to the baseline.

Orientations	Method	TE	RE	Time Cost (s)
L	YANG's	1.53	/	18.78
	Our	0.82	0.0050	9.22
R	YANG's	1.52	/	18.80
	Our	0.82	0.0050	9.23
F	YANG's	1.48	/	18.85
	Our	0.80	0.0048	9.22
U	YANG's	1.53	/	18.79
	Our	0.81	0.0049	9.24
D	YANG's	1.55	/	18.83
	Our	0.83	0.0052	9.26

"F", "L", "R", "U", and "D" represent front, left, right, upwards, and downwards orientations. YANG's method served as the baseline.

This paper conducted several ablation experiments, to verify the impact of different backbones and the addition of positional encoding on the model's prediction of the translational offset error and the rotational pose offset error. ResNet50, ResNet18, ResNet101, VGG16, VGG19, DenseNet-121, and DenseNet-201 were selected as the backbone feature extraction networks. For each backbone, experiments were conducted with and without positional encoding. The experimental results, shown in Table 5 and Figure 8, indicate that different backbones and the addition of positional encoding significantly affect the

model's final prediction accuracy. The version of the model with ResNet50 as the backbone feature extraction network and positional encoding added achieved the best performance in predicting the translational offset error and the rotational pose offset error, reaching 8.1 mm and 0.0049, respectively.

Table 5. The impact of different backbones within the model on the accuracy of predicting the translational offset error and the rotational pose offset error.

Backbones	Positional Encoding	TE	RE
ResNet50	✓	0.81	0.0049
	✗	1.19	0.0051
ResNet18	✓	0.92	0.0053
	✗	1.33	0.0054
ResNet101	✓	0.86	0.0050
	✗	1.21	0.0050
VGG16	✓	0.96	0.0053
	✗	1.32	0.0055
VGG19	✓	0.99	0.0054
	✗	1.33	0.0055
DenseNet-121	✓	0.92	0.0054
	✗	1.23	0.0055
DenseNet-201	✓	1.10	0.0055
	✗	1.38	0.0057

A check mark (✓) indicates that the model included positional encoding, while a cross (✗) indicates that the model did not include positional encoding.

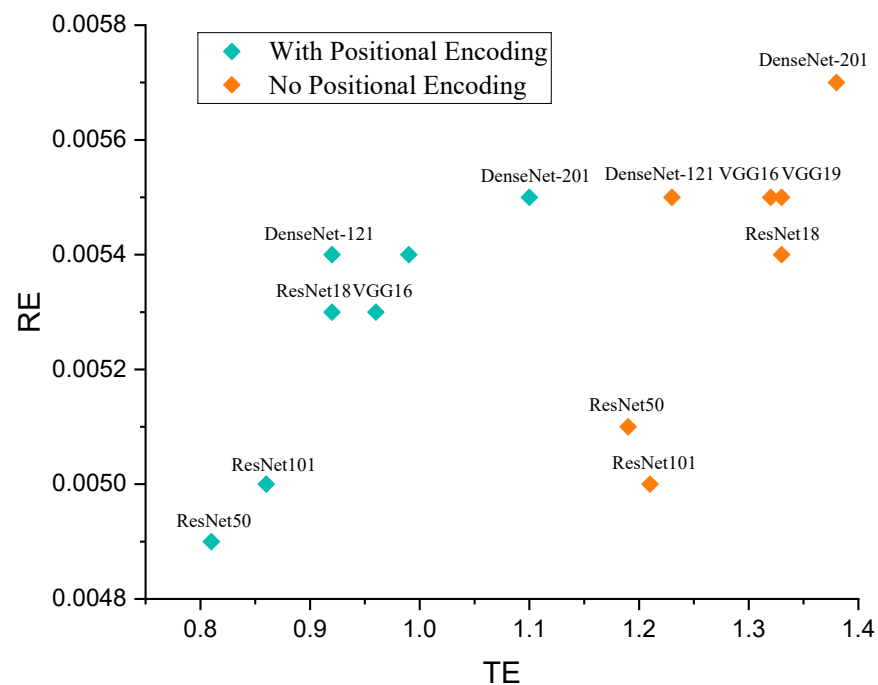


Figure 8. Distribution of translational and rotational error for different model backbones with and without the addition of positional encoding.

5. Discussion

This work proposes a transformer-based approach that achieves end-to-end prediction of translational and rotational errors between the robotic pollination arm's end effector

and the target pollination position, solely using RGB image information. Our experimental results demonstrate that this method effectively narrows the error range of the robotic end effector within a known error margin, thereby enhancing the overall efficiency of the robotic pollination.

In this work, our method exhibited slight variations in results when dealing with flowers of different orientations, particularly larger translational and rotational errors with downwards-facing flowers. This may be attributed to the relative positioning of the robot's camera angle to the flower orientation, complicating the accurate recognition and localization of flowers in specific orientations. Through experimentation, we also observed that the versions of the model that utilized positional encoding performed better in accurately predicting translational and rotational errors. Additionally, the different backbone networks used for extracting features from input images had a significant impact on the model's performance.

This work has potential limitations despite the demonstrated effectiveness of the proposed method in reducing translational and rotational errors between the pollination robot's end effector and the target: (1) The dataset used in this study was specifically collected for experimental purposes, which may limit the model's generalization capabilities across varied environments and flower types; (2) The model's performance may be compromised under different lighting conditions and in obstructed environments, impacting its overall effectiveness. These issues emphasize the need for further enhancements before practical deployment, necessitating future research to explore additional methods that improve the model's adaptability and reliability in diverse agricultural environments.

6. Conclusions

This work presents an innovative method that utilizes the powerful spatial learning and understanding capabilities of a transformer-based deep learning model to achieve end-to-end prediction of translational and rotational errors between the pollination robot's end effector and the target pollination position using only RGB images. Our experimental results demonstrate that this method is effective in further reducing the error range of the pollination robot's end effector within a known error range, thereby improving the overall efficiency of the pollination robot. Future work could focus on investigating the prediction of translational and rotational errors between the robotic end effector and target positions within datasets containing a more diverse range of flower types, under varying lighting conditions and occlusions. Such work would aim to enhance the robustness and generalization capabilities of the model, providing a feasible approach to improving the accuracy of generic robotic end effectors.

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Abbreviations

The following abbreviations are used in this manuscript:

TE translational error
RE rotational error

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